

WEEK 5 STATUS REPORT: ATTACK SIMULATION & FAULT-TOLERANCE PIPELINES

1. Disruption & Outage Simulation

- 1 Executed an administrative interface shutdown on Router 2 (R2) via its physical GigabitEthernet0/0 interface link.
- 2 This effectively simulated an offline node condition, mimicking an architectural hardware failure or network infrastructure drop.

2. Exception Isolation and Code Resilience

- 1 Restructured the primary script architecture loop using explicit python **try/except** parameter matrices.
- 2 Isolated '**socket.timeout**' exceptions to handle completely unresponsive destinations smoothly.
- 3 Isolated '**paramiko.AuthenticationException**' constraints to handle credential mismatch anomalies.
- 4 Verified that when target R2 hits a connection failure block, the script logs the specific error message to console instead of throwing an unhandled trace crash, safely shifting focus to successfully process remaining healthy network targets.